PROJECT DEVELOPMENT PHASE:

1. Data Layer Development (The Foundation)

Before the flow can run, the "buckets" for the data must exist.

- **Table Creation:** Create the `u_book` table.
  - *Fields:* Title, Author, ISBN (String), Status (Choice: Available, Borrowed, Reserved).
- **Relationship Building:** Create the `u_borrow_request` table.
  - *Reference Fields:* Link this table to the `sys_user` table (for the Borrower) and the `u_book` table (for the Book).
- **Data Import:** Use **Import Sets** or **Data Foundations** to upload your initial library catalog from an Excel/CSV file into the `u_book` table.

2. Logic Layer Development (Flow Designer)

This is the "brain" of your system that you were working on in your screenshot.

The "Borrow Request" Flow:

1. **Trigger:** Set to `Created` on the `u_borrow_request` table.
2. **Action - Ask for Approval:** * **Record:** Drag the `u_borrow_request` record pill here.
   - **Rules:** Set to "Anyone Approves" $\rightarrow$ **Librarian Group**.
3. **Flow Logic - If/Else:**
   - **If Approved:** * Action: **Update Record** (`u_book`) $\rightarrow$ Set *Status* to "Borrowed".
     - Action: **Send Email** to the User (Template: "Book Ready for Pickup").
   - **If Rejected:**
     - Action: **Update Record** (`u_borrow_request`) $\rightarrow$ Set *State* to "Closed Incomplete".
     - Action: **Send Email** to the User (Template: "Request Denied").

3. UI & Experience Development

How the users and librarians see the system.

- **Catalog Item:** Create a "Borrow a Book" item in the **Service Catalog**.
  - Use a **Reference Qualifier** so users can only select books where `Status = Available`.
- **Form Layouts:** Use **Form Designer** to keep the librarian's view clean. Group "Borrower Info" and "Book Details" into separate sections.
- **Workspace/Dashboard:** Build a **Librarian Dashboard** using "Platform Analytics" to show:
  - *Total Books Out.*
  - *Pending Approvals* (using a List Widget).
  - *Overdue Books.*

4. Coding & Scripting (Advanced)

Sometimes "Low-Code" isn't enough.

- **Client Scripts:** Use these to auto-fill the user's Department when they open the request form.
- **Business Rules:** Create a "Display" Business Rule to prevent a user from requesting a second book if they already have an overdue one.
- **Due Date Calculation:**

  $$Due\ Date = gs.daysAgo(-14)$$

  (Use a Script Action in the flow to set the return date exactly two weeks from today.)

5. Error Handling & Unit Testing

- **The Error Handler:** Use the "Error Handler" switch (visible in your screenshot) to catch issues. If a librarian's email is missing, the flow should log an error rather than crashing.
- **Unit Tests:** Manually "Test" the flow inside Workflow Studio using different user records to ensure the logic branches correctly.